Fiscal Year:	FY 2009	**Task Last Updated:**	FY 09/16/2009
PI Name:	Ronca, April Elizabeth Ph.D.		
Project Title:	Hypergravity Effects On The Maternal-Fetal System		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Musculoskeletal Biology (4) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	april.e.ronca-1@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650.400.6019
Organization Name:	NASA Ames Research Center		
PI Address 1:	Space Biosciences Research Branch		
PI Address 2:	MS 261		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:	November 2019: PI is located at NASA Ames Research Center and remains affiliated with Wake Forest University School of Medicine		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	12	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 12/31/2008, per A. Chu/ARC (Ed., 8/08) NOTE: Received NCE to 8/31/2008 per A. Chu (Ed., 8/07) 8/06--funding reinstated through 8/31/2007, per J. Cohen (HQ). Retaining original start date of 12/01/2002.		
Key Personnel Changes/Previous PI:	0		
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Task Description:	This project will advance our understanding of gravity's influence(s) on mammalian pregnancy, parturition and offspring development and their inter-relations. Through a series of integrative experiments, we seek to uncover similarities between prenatal exposure to increased (or hyper-) gravity and other intrauterine events, such as prenatal nutrients and oxygen that 'program', or set the stage for developmental outcomes and adult disease. The fetal programming hypothesis states that prenatal events set into motion a series of physiological alterations in the developmental program that may translate into adult disorders, in particular, cardiovascular disease, diabetes and mental illness. In the current project, Norway rats will be exposed to 2-g or 1.5-g hypergravity to establish continuous centrifugation as a model for fetal programming of adult disease. Non-pregnant females will be adapted to hypergravity prior to mating to eliminate effects of the adaptation process on fetal offspring. We have previously established reduced birth weights of rats exposed to 2-g or 1.5-g centrifugation during prenatal development. We have also observed decreased numbers of labor contractions in dams that underwent pregnancy during 2-g or 1.5-g centrifugation. Behavioral, physiological and molecular methods will be combined to ascertain mechanisms underlying effects of centrifugation on the maternal-fetal system. We will characterize and compare effects of hypergravity on uterine contractile activity during gestation and birth using intrauterine pressure measurement, hormonal and behavioral measures. Studies of fetal offspring at the time of birth will control exposure to uterine contractions by either administering specific numbers of artificially-produced contractions or by completely eliminating exposure to them. Comparisons will be made between offspring that undergo gestation in hypergravity and those exposed to bilateral occlusion of the uterine arteries during late pregnancy, the latter being an established model of fetal programming. Labor contractions will be related to neonatal variables, namely postpartum breathing and oxygenation, suckling and metabolic activity. Molecular analyses will focus on the expression of specific genes that are reactive to sensory simulation, the early, immediate genes, c-fos and c-jun in brain and adrenal of the newborn rats associated with birth. Longitudinal studies of the offspring will focus on adult cardiovascular responses to stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The newly emerging field of fetal programming, the idea that perinatal events direct or 'program' adult health and disease states, is uncovering startling insights into relationships between intrauterine events and adult health problems, including cardiovascular disease, diabetes and mental health disorders. Numerous epidemiological and animal studies provide unequivocal evidence for a strong association between low birth weight and adult cardiovascular disease. The goal of the proposed research is to determine whether hypergravity exposure of developing offspring is an appropriate model for fetal programming and can be used to further scientific understanding of fetal programming of adult disease.
	In recent years, the numbers of female astronauts in U.S. and international space programs has grown immensely. Women are expected to increase their participation in the NASA missions as the Vision for Space Exploration unfolds. This initiative calls for establishing an extended human presence on the moon and preparing for the Mission to Mars, requiring sustained habitation of the space environment by men and women alike. Within the private sector, there is a growing emphasis on space tourism with flights into low earth orbit and new initiatives for marketing space vacations. These unprecedented opportunities for expanded space exploration and habitation demand that we improve our understanding of both acute and enduring effects of space travel on women and the female reproductive system. It is vital that we identify potential reproductive risks for women astronauts and space tourists of tomorrow, and that we develop some understanding of reproduction in space before the inevitable first space conception transpires. Because there is almost no knowledge base on the effects of the space environment on the female reproductive system, ground-based animal studies comprise important first steps. Allowing female astronauts to serve as the 'guinea pigs' for uncovering space-related changes in reproductive physiology and function, particularly for long-duration missions, poses a risky approach to the problem. Our productivity provides continuing evidence of our excellent track record for timely and regular dissemination of our findings throughout the scientific community as well as strong media interest. Three invited book chapters on reproduction and space have appeared in edited volumes (Advances in Space Biology and Medicine and Sourcebook of Model Organisms in the Biomedical Sciences). Nineteen empirical papers have appeared in scientific journals and 41 published abstracts were presented at scientific meetings. Our research was prominently featured in 11 invited national and international colloquia given by the PI. The documentary we filmed at NASA Ames Research Center in 2004 entitled 'Science at the Edge: Breaking Nature's Rules' has been nationally and internationally aired on The Discovery Channel since 2005. In this documentary, the PI was interviewed and the importance of these centrifuge experiments was discussed. The PI was interviewed by the History Channel as part of a Universe series segment on 'Sex in Space' that aired in December 2008. Typically, fertility cycles of female astronauts are suppressed, so there have been no studies of human female reproductive processes under spaceflight conditions. However, environmental conditions are known to exert several adverse effects on the ovaries and on the hypothalamic-pituitary axis that regulates the ovulatory cycle and is highly susceptible to environmental factors. Stress alone can lead to ovulatory failure. Our recent experiments have uncovered adverse effects of altered gravity on critical phases of female reproductive processes related to estrus cycling and conception. These changes occur even when females are adapted to the altered gravity environment for over a week prior to mating. Collectively, our research suggests that extended exposure to 2-g centrifugation constitutes a chronic stressor that exerts acute, and possibly persistent, adverse effects on ovulatory function and the hypothalamic-pituitary axis regulating reproductive function. We have completed the key analyses required to confirm our findings and begin to uncover the underlying mechanisms. Female mammals undergo characteristic hormonal and cellular changes during the reproductive cycle. In rats, the estrous cycle averages 4-5 days in length and consists of four identifiable phases: proestrus, estrus, metestrus, and diestrus. Numerous environmental factors can interrupt the estrous cycle, such as stress, sterile mating, mechanical stimulation of the cervix, or continuous illumination. It was previously reported that, when centrifugation was initiated during diestrus, the length of the rat's estrous cycle increased due to prolonged diestrus. Female rats exposed to hindimb unloading to mimic aspects of microgravity exposure showed estrous cycle lengthening due to prolongation of the diestrus phase of the cycle. In this study, we hypothesized that continuous, long-duration (2 mo) exposure to 2-g centrifugation would alter patterns of estrous cycling and delay conception. The NASA Ames Research Center 24-foot Centrifuge was used to produce 2-g. Sexually mature female and male rats weight-matched then assigned to either 2-g centrifuge or non-centrifuge (stationary 1-g control) conditions, and to non-breeding or breeding conditions. In non-breeding females, estrous cycles were analyzed by examining vaginal

Task Progress:

cytology before and for 35 days during centrifugation. Females in the breeding condition were time-mated following 7 days of adaptation to centrifugation. Following adaptation to centrifugation, both centrifuged and non-centrifuged breeding females conceived, however centrifuged females took four-times longer than controls to achieve conception. Our results indicate that conception in 2-g adapted female rats was significantly delayed and resulted in overall diminished pregnancy outcomes.

Birth during centrifugation, even at modest g-loads of 1.5- to 2.0-g, is associated with increased rates of infant mortality, up to 60% mortality at 2.0-g. As a developmental biologist with expertise in studies of mammalian development, it was clear to the PI that existing paradigms for studying prenatal and neonatal development in centrifugation were lacking. Therefore, it was necessary for us to establish a feasible paradigm for centrifugation studies of pregnant rat dams and litters that would yield clear, non-confounded experimental data on live offspring that would be suitable for both ground-based and space platforms using ISS centrifuge. To facilitate comparisons with existing flight data, we exposed pregnant rats to centrifugation beginning on G11 of the rats 22-day gestation period, the timetable followed for the NIH.R2 spaceflight experiment. We selected a modest gravity level of 1.5-g tha enabled us to maximize neonatal survival while observing centrifugation effects on neonates. Comparisons were made among Centrifugation-exposed (HG), Stationary Yoked Control (SYC), Rotational Control (RC) and Vivarium (VIV) groups with dams weight-matched across conditions. We outfitted the centrifuge with high-resolution CCD cameras to allow for collection of behavioral data and monitoring of the dams and offspring during centrifugation. We applied conventional statistical procedures for developmental studies in which each litter is considered to be a single unit. Briefly, multiple offspring from the same dam do not represent repeated, independent observations because the littermates share in common a dam, a genetic substrate, and a host of environmental variables that are not shared with offspring from another litter. Since littermates are non-independent observations, offspring from a given dam represent only one observation. In addition, we anticipated some neonatal mortality, therefore it was important to control for litter size across the HG and SYC conditions. We did this by matching size and composition of the SYC litters to HG litters on a daily basis. The RC and VIV groups remained undisturbed throughout the study, the latter providing a control for shrinking litter size in the HG group. This paradigm has yielded important advances in the quality and interpretability of the data that we can derive from animals developing during centrifugation. Following centrifugation onset, HG dams showed an immediate decline in body mass. Food intake and water intake (per 100gm dam body mass) were reduced in HG relative to SYC dams during the first three days of centrifugation but not thereafter. Behavioral activity of the HG and SYC dams encoded from timelapse video (12:1 tape:playback ratio) revealed that HG dams were consistently less active than SYC dams until late in gestation. Non-pregnant animals show a similar pattern of adaptation coupled with increases in corticosterone (CORT) that persist for only four to six days following the onset of centrifugation.

To ascertain the role of maternal reproductive experience in neonatal mortality in centrifugation, compared mothers of differing parities. Primigravid (first pregnancy) and bigravid (second pregnancy) female rats were exposed to 1.5-g centrifugation from G11 throughout birth and the first postnatal week. On the day of birth, litter sizes were identical across gravity and parity conditions although significantly fewer live neonates were observed among centrifugation-reared litters born to primigravid dams as compared to bigravid dams (82% and 94%, respectively; 1.0-g controls, 99%). Birth weights of the centrifugation-reared neonates were significantly less than those of controls (SYC, 6.9 +/- 0.4 gm; HG, 6.0 +/- 0.6 gm). Fetal mortality did not differ across gravity conditions, however disruption of pup-directed maternal care was highly correlated with neonatal mortality ($R2 = 0.99$, $p < 0.02$). Prior pregnancy and birth can ameliorate neonatal losses in centrifugation during the first 24 hrs after birth. In light of the known linkage between birth weight and adult disease, one of the most striking observations of this study is that, whereas the body weights of bigravid and primigravid dams exposed to centrifugation were vastly different one another (Primigravid, 252.1 +/- 11.4 gm; Bigravid, 302.4 +/- 22.6 gm) birth weights of the offspring did not differ across parity conditions.

In a separate study, we exposed rat dams to continuous centrifugation (20 RPM) at one of three g-loads (1.5, 1.75, or 2.0-g) from G11 until birth. Comparisons were made with Rotational Control (1.03-g) and Stationary Control (1.0-g) conditions. Pup body weights in each gravity condition differed from every other gravity condition. Although birth weights of neonates in the RC condition were suppressed relative to those in the SC condition, possibly due to the stress of centrifugation or changes in uterine blood flow at different g-loads, increased gravity resulted in greater decreases in birth weight. Together these findings indicate that centrifugation exposure during the latter half of gestation is associated with reduced birth weight. This outcome is independent of maternal body weight, age and parity.

We have a long-standing interest in the adaptive effects of labor contractions in the newborn. Labor in mammals facilitates the transition from fetus to newborn. In response to the stress of powerful contractions, the fetus secretes copious amounts of adrenaline and noradrenaline from adrenal and extra-adrenal tissues. Catecholamine release assists the neonate’s survival outside of the womb by affording protection against hypoxia and exerts a number of positive effects on respiratory variables, including lung liquid absorption, surfactant release, and increased lung compliance. There also exists a positive relationship between vaginal birth and neonatal outcome. Vaginal delivery, relative to elective cesarean birth, is associated with increased neonatal metabolic rate, enhanced blood flow to vital organs and improved neurodevelopmental status. We have been able to analyze neonatal adaptation to the extrauterine environment using a unique procedure we devised to 'build' a birth. Using this technique, we can apply the major components of birth to fetal rats at term and otherwise perform a cesarean delivery. These components are: 1) Compression to simulate labor contractions, 2) Umbilical cord occlusion, 3) Cooling from intrauterine temperature (37.5 deg C) to room temperature (21 deg C), and 4) Stroking to mimic postpartum licking by the dam. In a series of studies in which we have applied these birth components singly and in combination, we have found that simulated labor contractions are essential for the maintenance of breathing at birth; without contractions or some form of physical stimulation, the infants will not survive. We have recently observed decreases in pH, pO2, and oxygen saturation in animals deprived of labor contractions. We have also shown that labor contractions also facilitate the onset of suckling at birth. These changes in rats, like humans, are associated with catecholamine release. There is known to be widespread neural expression of the immediate early gene, c-fos in newborn mice. We have hypothesized that the brain is 'turned on' at birth by labor contractions and have measured c-fos expression throughout the brain (with special attention to the locus coeruleus and other areas involved in arousal) in rats exposed to and deprived of labor contractions.

In the NASA NIH.R2 spaceflight mission, pregnant rats were flown on the Space Shuttle for 9 days beginning on Gestational day (G) 11 of their 22-day pregnancy until G20, approximately 48 to 72 hrs prior to birth. The NIH.R2 dams successfully completed their pregnancies although twice as many labor contractions were observed in the Flight dams relative to ground controls. These results replicated our findings from NIH.R1 in which pregnant dams were launched on G11 and landed on G20 of gestation. We repeated the NIH.R2 study in the 1.5-g centrifugation environment.

These findings have led us to hypothesize that a given amount of intrauterine force is required to expel the fetuses from

the womb during parturition. Following exposure to hypogravity, more contractions may be needed to complete parturition because each contraction is less forceful than normal. This observation is supported by the finding that connexin-43 (but not connexin-26) proteins were decreased in the myometrium of spaceflight dams. Connexin-43 is a key step in the proliferation of gap junctions that participate in uterine contractions during labor. In the present application, we will consider the role of oxytocin as well as connexin proteins as a mechanism for centrifugation-induced changes in uterine contractility. We have utilized a telemetric device for measuring IUP. This technique enabled us to make precise measurements of intrauterine pressure in behaving dams and thereby ascertain the force of contractions in altered gravity. Importantly, the wireless sensor will be suitable for making IUP measurements in freely moving pregnant rats onboard future spaceflight vehicles or the ISS In view of the pervasive effects of labor contractions on the newborn's adaptation to life outside of the womb, and the effects of altered gravity on uterine contractions, we hypothesized that uterine activity is one component of fetal programming. Our results support this hypothesis. The third major aim of our work is to further our understanding of altered gravity as a chronic stressor in females. Stress induced by altered gravity may underlie delayed conception. Previously, our work has incorporated a contemporary perspective in maternal-fetal medicine: ‘Intrauterine programming’ is revealing astounding insights into the prenatal origins of adult disease arising from environmental factors. Adult disease states that have been implicated include metabolic, endocrine, and cardiovascular pathophysiologies. Importantly, our results indicate that 2-g centrifugation beginning prior to conception leads to reduced body weights of neonatal offspring, increased body weights and hormonal alterations in adulthood even when animals are reared at 1-g. These findings are helping to establish 2-g centrifugation as a unique model for chronic stress that induces programming effects on progeny reared at 1-g. Mere exposure of the female reproductive system (prior to conception) may be sufficient to induce persistent changes in the health and well-being of offspring. Additional hormonal and neural studies are needed to clarify effects of altered gravity as a chronic stressor, including both acute and persistent effects on the female reproductive system, and to identify meaningful countermeasures for use in female astronauts. These concerns are vital to our future space endeavors.

In summary, our work has provided important analyses of altered gravity effects on reproductive physiology of the female mammal and influences on developing offspring. The Vision for Space Exploration calls for conducting space biology studies on the astronauts themselves in future flights rather than relying on animal models to ascertain the physiological consequences of space travel. However is it vital that we understand in advance of long-term habitation of space by female astronauts the requisite changes in reproductive function, whether these changes persist after return to Earth, and mechanisms of cross-generational transmission. Collectively, these animal studies are vital to understanding the experience of astronauts during long-duration missions and persistent effects of spaceflight on reproductive processes, including how altered gravity affects female reproductive processes and the specific underlying mechanisms. Continuing attention to these types of animal studies is sorely needed for developing meaningful countermeasures for female astronauts that undertake long-duration missions into space, and for understanding potential consequences of reproduction in space.

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